

Boyao Liao | Curriculum Vitae

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RESEARCH INTERESTS

Learning theory, deep learning theory, optimization theory.

EDUCATION

Jinan University – University of Birmingham Joint Institute **Guangzhou, China & Birmingham, UK**
B.Sc. in Mathematics and Applied Mathematics *September 2023–Present*

- GPA: 4.29/5, rank: 3/68, average score: 92.9/100.
- Coursework: Integer Programming and Combinatorial Optimization (97/100), Real Analysis (94/100), Data Structure (96/100), Probability and Statistics (98/100), Linear Algebra and Linear Programming (97/100) and more.

SELECTED AWARDS & HONORS

- **Best Student Paper Award**, The 37th International Conference on Algorithmic Learning Theory (ALT), 2026
- **National Scholarship**, Ministry of Education of China, 2024
- **Outstanding Student**, Jinan University, 2024

RESEARCH EXPERIENCE

Research Intern, Dartmouth College *2025–Present*
Advisor: Prof. Yaoqing Yang

Project: Suspicious Alignment of SGD – Fine-Grained Step-Size Analysis (Best Student Paper Award for ALT 2026)

- Contributed technical lemmas and auxiliary proofs for the adaptive critical step-size analysis.
- Proved auxiliary results distinguishing the alignment-decreasing and alignment-increasing regimes.
- Conducted and analyzed experiments supporting the theoretical characterization of SGD alignment dynamics.

Project: Depth, Not Data – Hessian Spectral Bifurcation (ISIT 2026)

- Developed the proof framework for the Hessian spectral analysis of deep linear networks.
- Proved that the dominant and the bulk Hessian eigenvalue ratio equals the network depth $\Theta(L)$ under exactly aligned and balanced initialization.
- Derived a sufficient condition for bifurcation.

Research Intern, City University of Hong Kong *2025–Present*
Advisor: Prof. Hongyi Jiang, City University of Hong Kong

Project: Online Shortest-Path Routing under Anytime Budget Constraints (Under Review at KDD 2027)

- Led the development of the project and designed an algorithm for online shortest-path routing under hard anytime budget constraints in a combinatorial semi-bandit setting.
- Developed the theoretical analysis and proved the main regret upper bound for the proposed algorithm.

SELECTED PUBLICATIONS

- **Suspicious Alignment of SGD: A Fine-Grained Step Size Condition Analysis**
Shenyang Deng, **Boyao Liao**, Zhuoli Ouyang, Tianyu Pang, Minhak Song, Yaoqing Yang
The 37th International Conference on Algorithmic Learning Theory (ALT), 2026. Best Student Paper Award.
arXiv: 2601.11789
- **Depth, Not Data: An Analysis of Hessian Spectral Bifurcation**

Shenyang Deng, **Boyao Liao**, Zhuoli Ouyang, Tianyu Pang, Yaoqing Yang
IEEE International Symposium on Information Theory (ISIT), 2026.
arXiv: 2602.00545

MANUSCRIPTS UNDER REVIEW

- **Online Shortest-Path Routing under Anytime Budget Constraints**

Boyao Liao, Hongyi Jiang

Under Review at 33rd SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2027.

WORK IN PROGRESS

- **Extension of Depth, Not Data: An Analysis of Hessian Spectral Bifurcation.** *Manuscript in preparation for submission to IEEE Transactions on Information Theory (TIT).*

Extended the dominant/bulk Hessian spectral analysis beyond exactly aligned and balanced initialization.

- **Extension of Online Shortest-Path Routing under Anytime Budget Constraints.** *Manuscript in preparation for submission to Mathematics of Operations Research (MOR).*

Developed a new algorithm for the multi-resource setting and established matching upper and lower regret bounds.

ACADEMIC SERVICE

- Reviewer, NeurIPS 2026.

TALK

- Time: September 28, 2026 (to be given)

Location: Dartmouth–Berkeley–Rice AI Reading Group

Topic: Depth, Not Data – Hessian Spectral Bifurcation

SKILLS

Programming Python, MATLAB, C, R.

ML/Research PyTorch distributed libraries (DDP, FSDP, torch.distributed), Probabilistic DL libraries (e.g., Pyro), Various open-source repositories.

Writing/Tools L^AT_EX, Markdown, Various scientific plotting tools.